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(1) Research Program/Project/Study Title: GINGER ICE CREAM	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:		
Component Leader:		
Academic Rank:		

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Pilar
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity	(8) Discipline/Field of Specialization
(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) (9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)	
(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) Ice cream (iced cream or cream ice) is a sweetened frozen food typically eaten as a snack or dessert. It is usually made from dairy product, such as milk and cream, and often combined with fruits or other ingredients and flavors. It is typically sweetened with sugar or sugar substitutes. Typically, flavorings and colorings are added in addition to stabilizers. Ginger is a common ingredient in Asian and Indian cuisine. However, ginger has been used for its medicinal properties for centuries among many cultures. Ginger has a long history of use for relieving digestive problems such as nausea, loss of appetite, motion sickness and pain. Many studies have suggested that increasing consumption of plant foods like ginger decreases the risk of obesity, diabetes, heart disease and overall mortality while promoting a healthy complexion and hair, increased energy and overall lower weight. Philippines is a grower of ginger. In the province of Capiz, the upland areas are producing tons and tons of ginger and sold whole during market days. Some entrepreneurs processed it into instant powder and sold by a kilo. Since gingers is really abundant in our place and have medicinal benefits to the person with and without sickness, the researcher is encouraged to produce a product where everybody can have it. The present study was developed to determine the acceptability of Ginger Ice Cream. Specifically it sought: to determine the acceptability of ginger ice cream in terms of appearance, color, aroma, flavor, and general acceptability; to determine the significant difference in the acceptability of Ginger Ice Cream in terms of appearance, color, aroma, flavor, and general acceptability; and to determine the cost and yield of the product.	
(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)	

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This is an experimental research which utilized 30 panelists purposively selected to validate the product. The study used a single factor experiment in completely randomized design (CRD) in 5 treatments and 3 replications.

Treatment A - 3.3% Ginger

Treatment B - 3.85% Ginger

Treatment C - 4.4% Ginger

Treatment D – 4.95% Ginger

Treatment E – 0% Ginger

Material / Informants

The formulation of the product consists of Whip cream 39%, condensed milk 29%, chocolate 11%, ginger powder 11% and others consists of 11%. The tools / equipment needed are: measuring cup, grammer, weighing scale, large bowl, wire whip, and plastic cups.

The general preparation of the materials:

Steps in making Ginger Powder

1. Place ginger root in a bowl and soak in water for 3-5 minutes and wash nicely under running water to remove the dirt completely.
2. Cut into thin pieces using a sharp knife, the faster drying out. Once the ginger cuts dried completely, powder it finely in a dry grinder, put it in a food processor, if there is any bits remaining grind again finely, then sieve to eliminate coarser particles. When done, store in a dry, airtight glass container.

Procedures on How to Make the Ginger Ice Cream

Place heavy cream on the mixing bowl, add condensed milk, vanilla and ginger. Beat thoroughly to its peak. Add the chocolate for additional flavor blend again thoroughly. When done pour the mixture in cups and place in the refrigerator.

Analysis/Procedure

The acceptability of the product was evaluated and sensory qualities of the product were determined using the score card as to their appearance, color, aroma, flavor, and general acceptability. Discussion regarding the process of the evaluation was given emphases for the validity of the results.

Sensory Evaluation

The sensory evaluation was carried out on ice cream samples after 18 hours of hardening. The coded samples were served to panelist during morning or normal daylight. The panel consisted of thirty 30 panelists who are faculty and students of CapSU, Pilar. The hardened ice cream samples were placed in a sealed thermal box to maintain a temperature of approximately 10 °C. The samples were

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evaluated for attributes of appearance, body and texture, flavour, mouth-feel and overall acceptability scores on a 9-point Hedonic Scale.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Appearance

Table 1 presents the acceptability of ginger ice cream as evaluated by the panel of consumers in terms of appearance. It shows that the Treatment B ($M=8.20$, $SD=0.501$) and E ($M=8.30$, $SD=0.483$). was extremely liked by the respondents who evaluated the ginger ice cream in terms of appearance. It was followed by the Treatment A ($M=7.73$, $SD=0.940$), Treatment C ($M=7.50$, $SD=0.527$) and Treatment D ($M=7.40$, $SD=0.733$). Based on the result, we can conclude that all treatments of ginger ice cream are acceptable in terms of appearance and liked very much by the panel of consumers who evaluated the said product.

Table 1. Level of Acceptability in terms of Appearance.

TREATMENTS	<i>M</i>	<i>SD</i>	VERBAL INTERPRETATION
Treatment A	7.73	0.940	LIKE VERY MUCH
Treatment B	8.20	0.501	LIKE EXTREMELY
Treatment C	7.50	0.527	LIKE VERY MUCH
Treatment D	7.40	0.733	LIKE VERY MUCH
Treatment E	8.30	0.483	LIKE EXTREMELY
TOTAL	7.83	0.732	LIKE VERY MUCH

Color

Table 2 presents the acceptability of the ginger ice cream as tested/evaluated by the panel of consumers in terms of color. In terms of color it shows that treatment B is the most acceptable is liked extremely by the participants who evaluated the ginger ice cream ($M=8.20$, $SD=0.391$), followed by treatment E ($M=8.10$, $SD=0.817$) which is considered as acceptable and is liked very much, then treatment A ($M=8.07$, $SD=0.562$) which is also considered as acceptable, same with treatment C ($M=7.70$, $SD=0.457$) and lastly treatment D ($M=7.37$, $SD=0.428$) which is still acceptable and was liked very much. Based on the result it can taken that all ginger ice cream treatments are all acceptable in terms of color and like very much by the panel eof consumers who tested the said product.

Table 2. Level of Acceptability in terms of Color.

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TREATMENTS	<i>M</i>	<i>SD</i>	VERBAL INTERPRETATION
Treatment A	8.07	0.562	LIKE VERY MUCH
Treatment B	8.20	0.391	LIKE EXTREMELY
Treatment C	7.70	0.457	LIKE VERY MUCH
Treatment D	7.37	0.428	LIKE VERY MUCH
Treatment E	8.10	0.817	LIKE VERY MUCH
TOTAL	7.89	0.616	LIKE VERY MUCH

Aroma

Table 3 presents the acceptability of the ginger ice cream as tested/evaluated by the panel of consumers in terms of aroma. It shows that treatment B is the most acceptable is liked extremely by the participants who evaluated the ginger ice cream ($M=8.23$, $SD=0.417$), followed by treatment E ($M=8.10$, $SD=0.353$) which is considered as acceptable and is liked very much, then treatment A ($M=7.90$, $SD=0.522$), treatment C ($M=7.50$, $SD=0.549$) and treatment D ($M=7.43$, $SD=0.353$) which is also considered both as acceptable and is also liked very much.

Based on the result it can be concluded that all ginger ice cream are acceptable in terms of texture and like very much by the panel of consumers who tested the said product

Table 3. Level of Acceptability in terms of Aroma

TREATMENTS	<i>M</i>	<i>SD</i>	VERBAL INTERPRETATION
Treatment A	7.90	0.522	LIKE VERY MUCH
Treatment B	8.23	0.417	LIKE EXTREMELY
Treatment C	7.50	0.549	LIKE VERY MUCH
Treatment D	7.43	0.353	LIKE VERY MUCH
Treatment E	8.10	0.353	LIKE VERY MUCH
TOTAL	7.83	0.536	LIKE VERY MUCH

Flavor

Table 4 presents the acceptability of the ginger ice cream as tested/evaluated by the panel of consumers in terms of flavor. It shows that treatment B ($M=8.20$, $SD=0.571$) and treatment E in terms of flavor ($M=8.20$, $SD=0.281$) are the most acceptable and is liked extremely by the participants who evaluated the ginger ice cream. It is followed by treatment A ($M=7.83$, $SD=0.503$) which is considered as acceptable and is liked very much, treatment C ($M=7.67$, $SD=0.609$) and lastly treatment D ($M=7.30$, $SD=0.399$) which is still acceptable is also liked very much.

Results revealed that ginger ice cream is acceptable in terms of flavor and like very much by the panel of consumers who tested the said product.

TREATMENTS	<i>M</i>	<i>SD</i>	VERBAL INTERPRETATION
Treatment A	7.83	0.503	LIKE VERY MUCH
Treatment B	8.20	0.571	LIKE EXTREMELY

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Treatment C	7.67	0.609	LIKE VERY MUCH
Treatment D	7.30	0.399	LIKE VERY MUCH
Treatment E	8.20	0.281	LIKE EXTREMELY
TOTAL	7.84	0.580	LIKE VERY MUCH

Table 4. Level of Acceptability in terms of Flavor.

Acceptability

Table 5 revealed the overall acceptability of the ginger ice cream as varied proportions as tested/evaluated by the panel of consumers. It shows that the ginger ice cream is highly acceptable with a mean of 7.86 and is reasonably interpreted as like very much as evaluated and tested by the panel of consumers. Specifically, treatment B ($M=8.20$, $SD=0.571$) and treatment C ($M=8.20$, $SD=0.421$) are the most acceptable and is liked extremely. It is then followed by treatment A ($M=7.97$, $SD=0.483$) which is considered as acceptable and is liked very much, then treatment C ($M=7.53$, $SD=0.525$) and lastly treatment D ($M=7.40$, $SD=0.438$) who got the lowest mean but is also considered acceptable and liked very much by the consumers.

Based from the result it can expressed that ginger ice cream is over all acceptable and liked very much by the panel of consumers who tested the said product

Table 5. Level of Acceptability in terms of Acceptability.

TREATMENTS	M	SD	VERBAL INTERPRETATION
Treatment A	7.97	0.483	LIKE VERY MUCH
Treatment B	8.20	0.571	LIKE EXTREMELY
Treatment C	7.53	0.525	LIKE VERY MUCH
Treatment D	7.40	0.438	LIKE VERY MUCH
Treatment E	8.20	0.421	LIKE EXTREMELY
TOTAL	7.86	0.579	LIKE VERY MUCH

Table 6 presents the analyzed result from the one-way ANOVA in the experimentation using varying proportions of ginger ice cream. The hypothesis stated that there is no significant difference between the ginger ice cream in terms of appearance, color, aroma, flavor and general acceptability was rejected due to the significant p -value 0.010, 0.008, 0.000, 0.001 and 0.001 respectively.

This means that there is a significant difference among the treatments of ginger ice cream.

Table 6. One-Way ANOVA (Experimental Group)

		Sum of Squares	df	Mean Square	F
Appearance	Between Groups	6.609	4	1.652	3.781
	Within Groups	19.667	45	.437	

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	Total	26.276	49			
Color	Between Groups	4.813	4	1.203	3.933	.008
	Within Groups	13.767	45	.306		
	Total	18.580	49			
Aroma	Between Groups	5.067	4	1.267	6.341	.000
	Within Groups	8.989	45	.200		
	Total	14.056	49			
Flavor	Between Groups	5.809	4	1.452	6.114	.001
	Within Groups	10.689	45	.238		
	Total	16.498	49			
Acceptability	Between Groups	5.609	4	1.402	5.813	.001
	Within Groups	10.856	45	.241		
	Total	16.464	49			

Note: * $p < 0.01$.

The data revealed as shown in table 7 presents the Post hoc analysis (p-values for pairwise t-Test) for the ginger ice cream in varying proportions in terms of appearance, color, aroma, flavor and general acceptability. According to Bonferroni, the result further affirms that there is a significant difference that exist among the treatments with varying proportions of ginger ice cream since the significant is less than the 0.01 alpha level.

In other words, all then treatments are different and acceptable regardless of appearance, color, aroma, flavor and acceptability.

Table 7. Post hoc analysis (p-values for pairwise t-Tests)

	APPEARANCE					COLOR					AROMA		
Treatment	T _A	T _B	T _C	T _D	T _E	T _A	T _B	T _C	T _D	T _E	T _A	T _B	T _C

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T _A	1.000	1.000	1.000	.616		1.000	1.000	.069	1.000		1.000	0.514	.241	1.000	
T _B	1.000		.223	.096	1.000	1.000		.492	.016	1.000	1.000		.006	.002	1.000
T _C	1.000	.223		1.000	.096	1.000	.492		1.000	1.000	.514	.006		1.000	.044
T _D	1.000	.096	1.000		.039	.069	.016	1.000		.048	.241	.002	1.000		.017
T _E	.616	1.000	.096	.039		1.000	1.000	1.000	.048		1.000	1.000	.044	.017	

Note: * $p < 0.01$.

Table 7. Post hoc analysis (p-values for pairwise t-Tests). Continuation

Treatment	FLAVOR					ACCEPTABILITY				
	T _A	T _B	T _C	T _D	T _E	T _A	T _B	T _C	T _D	T _E
T _A		.994	1.000	.184	.994		1.000	.547	.132	1.000
T _B	.994		.184	.002	1.000	1.000		.040	.007	1.000
T _C	1.000	.184		.994	.184	.547	.040		1.000	.000
T _D	.184	.002	.994		.002	.132	.007	1.000		.000
T _E	.994	1.000	.184	.002		1.000	1.000	.040	.007	

Note: * $p < 0.01$.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSIONS:

1. Ginger ice cream prepared with 3.85% ginger (Treatment B) obtained the highest acceptability in terms of appearance, color, flavor, aroma and general acceptability. However, all other treatments were liked very much by the panel of consumers who tested the said product.
2. There is a significant difference among the treatments of ginger ice cream. The result further affirms that there is a significant difference that exist among the treatments with varying proportions of ginger ice cream, in other words all the treatments are different and acceptable regardless of appearance, color, aroma, flavor and acceptability. Therefore the hypothesis

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which stated that there is no significant difference between the ginger ice cream in terms of appearance, color, aroma, flavor and general acceptability was rejected.

3. Treatment D prepared with 4.95% ginger composition obtained the highest cost among treatments.

RECOMMENDATIONS:

1. Ginger ice cream can be used to make ice cream.
2. The product maybe submitted for microbial analysis.
3. A similar study is to be conducted to test the shelf life of the product.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Ginger Ice Cream, IEC materials, Utility Model

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

Training Workshop on the Ginger Ice Cream development

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

Ali M.N., Prasad S. Studies on rheological properties of herbal ice cream prepared by selected medicinal herbs. *Journal of Multidisciplinary Research and Development* 2015; 2(3): 645-647

Misri Gozan , Muhammad Yusuf Arya Ramadhan, Andre Fahriz Perdana Harahap, Cut Nanda Sari, Yuswan Muharam, Widodo Wahyu Purwanto¹, Dewi Tristantini: Techno-Economic Analysis of Healthy Herbal Ice Cream Product *International Journal of Technology* 11(5) 931-940 (2020)